Linear Models Lecture 15: GMM Estimation and Efficiency

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Where We Left Off

In Lecture 14, we introduced IV estimation as a method of moments:

$$\frac{1}{n}\sum_{i=1}^{n} Z_i(Y_i - X_i'\beta) = 0$$

This works when the number of moment conditions ℓ equals the number of parameters k.

But what if we have more instruments than parameters?

- We cannot simultaneously set all moment conditions to zero
- We need a principled way to *get as close as possible*
- This is the **overidentification problem**

The Problem

Suppose you have $\ell > k$ moment conditions:

$$\mathbb{E}[g_i(\beta)] = 0, \qquad g_i \in \mathbb{R}^\ell, \quad \beta \in \mathbb{R}^k$$

In a sample, you cannot set all ℓ conditions to zero simultaneously—there are more equations than unknowns.

Example: You have three valid instruments for one endogenous variable. Each implies $\mathbb{E}[Z_{ji} e_i] = 0$. No single $\hat{\beta}$ satisfies all three exactly.

We need a principled way to get *as close as possible* to satisfying all of them. The choice of "close" will determine both the estimator and its efficiency.

Moment Equation Models

Let $g_i(\beta)$ be a known $\ell \times 1$ function of the ith observation and a $k \times 1$ parameter β.

Definition

A moment equation model *is defined by*

$$\mathbb{E}[g_i(\beta)] = 0$$

and a parameter space $\beta \in B$.

Identification requires $\ell \geq k$:

- $\ell = k$: **Just identified** — exactly enough information
- $\ell > k$: **Overidentified** — excess information (testable restrictions)
- $\ell < k$: **Underidentified** — insufficient information

Example: IV model: $g_i(\beta) = Z_i(Y_i - X_i'\beta)$

Everything Is a Moment Condition

Estimator	**Moment condition** $g_i(\beta)$	**MME**
Sample mean	$Y_i - \mu$	$\hat{\mu} = \bar{Y}$
OLS	$X_i(Y_i - X_i'\beta)$	$\hat{\beta} = (X'X)^{-1}X'Y$
IV	$Z_i(Y_i - X_i'\beta)$	$\hat{\beta} = (Z'X)^{-1}Z'Y$
2SLS	$Z_i(Y_i - X_i'\beta)$ with $W = (Z'Z)^{-1}$	$\hat{\beta}_{2SLS}$
GLS	$\bar{X}_i\Sigma^{-1}(Y_i - \bar{X}_i'\beta)$	$\hat{\beta}_{GLS}$

Key insight: All the estimators we have studied are *method of moments estimators* — they solve $\bar{g}_n(\hat{\beta}) = 0$. GMM generalizes to the overidentified case.

Where Do Moment Conditions Come From?

The table shows that every estimator *solves* moment conditions. But where do the conditions come from?

They come from **substantive claims about the world**—not from algebra.

A theory says something about what is exogenous, what is excludable, or where randomization occurs. That claim translates into an orthogonality condition $\mathbb{E}[g(W_i, \theta)] = 0$. The estimator follows.

The hard part is the translation. The same research question can lead to different moment conditions depending on what you are willing to assume.

We'll see this with two examples.

Example 1: Does Media Affect Political Attitudes?

Kern & Hainmueller (2009) study whether exposure to West German television reduced support for the East German regime.

The question: Does media exposure (X) affect regime support (Y)?

The problem: People who watch more television may differ from those who don't. If media consumption reflects pre-existing dissatisfaction, OLS confounds the effect of media with the effect of prior attitudes.

What moment condition should we write down? That depends on what we think is generating the variation in X.

Two Claims, Two Moment Conditions

Claim A: Media consumption is determined by geography (terrain blocks the signal). After controlling for observables, exposure is unrelated to unobserved attitudes.

$$\mathbb{E}[X_i\, e_i] = 0 \qquad \Longrightarrow \qquad \text{OLS}$$

Claim B: Even geographic variation is suspect—people in "no signal" areas may differ in other ways. But signal availability (Z) shifts exposure without directly affecting attitudes.

$$\mathbb{E}[Z_i\, e_i] = 0 \qquad \Longrightarrow \qquad \text{IV}$$

Kern & Hainmueller make Claim B. The instrument is geographic signal availability. The theory determines the moment condition. The moment condition determines the estimator.

Example 2: Does Winning Office Help You Win Again?

Lee (2008) studies the **incumbency advantage**: does holding office give you an electoral edge next time?

The problem: Winners and losers differ in quality, funding, and district partisanship. The moment condition $\mathbb{E}[D_i\, e_i] = 0$ (where D_i = won) is not credible.

The design: In very close elections, who wins is approximately random. A candidate who wins with 50.1% is comparable to one who loses with 49.9%.

This is a **local** claim: the theory doesn't say winning is random everywhere—only near the threshold. The moment condition must reflect this.

A Local Claim ⇒ A Local Moment Condition

In a regression discontinuity design, we estimate the treatment effect at the cutoff c using observations nearby.

Local linear regression solves a **kernel-weighted** moment condition:

$$\mathbb{E}\left[K_h(X_i - c) \cdot \begin{pmatrix} 1 \\ X_i - c \end{pmatrix}(Y_i - \alpha - \tau D_i - \beta(X_i - c))\right] = 0$$

where K_h is a kernel that downweights observations far from c.

This is recognizably a weighted OLS moment condition. But the **kernel weight encodes the theoretical claim**: only observations near the cutoff satisfy the "as good as random" condition.

The bandwidth h reflects how far from the cutoff you trust the continuity assumption. Wider h: more data, more precision, but the claim is more fragile. Narrower h: more credible, but noisier.

Three Claims, Three Moment Conditions

Theoretical claim	Moment condition	Estimator
Treatment is randomly assigned	$\mathbb{E}[D_i\, e_i] = 0$	OLS
Instrument is excludable	$\mathbb{E}[Z_i\, e_i] = 0$	IV / 2SLS
Local randomization at cutoff	$\mathbb{E}[K_h(X_i)\, D_i\, e_i] = 0$	Local linear RD

Each row reflects a *different* claim about where the exogenous variation comes from. Stronger claims (top) use more data. Weaker claims (bottom) use less, but are more credible.

The moment condition framework makes these tradeoffs explicit. When you have *more* moment conditions than parameters—as with multiple instruments—GMM tells you how to combine them optimally and how to test whether they are mutually consistent.

Getting as Close as Possible

Define the sample moment:

$$\bar{g}_n(\beta) = \frac{1}{n}\sum_{i=1}^{n} g_i(\beta)$$

When $\ell = k$: We can solve $\bar{g}_n(\hat{\beta}) = 0$ exactly.

When $\ell > k$: There are more equations than unknowns. In general, no β sets $\bar{g}_n(\beta) = 0$.

Intuition: Think of $\mu = Z'Y$, $G = Z'X$, and $\eta = \mu - G\beta$. We want η small. Regressing μ on G:

$$\tilde{\beta} = (G'G)^{-1}G'\mu$$

minimizes $\eta'\eta$. But we can do *better* with weighted least squares...

The GMM Criterion Function

For a positive definite $\ell \times \ell$ weight matrix W, the **GMM criterion function** is:

$$\boxed{J(\beta) = n\,\bar{g}_n(\beta)'W\,\bar{g}_n(\beta)}$$

- $J(\beta) \geq 0$ for all β (since $W > 0$)
- When $W = I_\ell$: $J(\beta) = n\|\bar{g}_n(\beta)\|^2$ (Euclidean distance)
- The factor n is for distributional convenience
- Different choices of W yield different estimators — and different efficiency

For the linear IV model:

$$J(\beta) = n(Z'Y - Z'X\beta)'W(Z'Y - Z'X\beta)$$

The GMM Estimator

Definition (GMM Estimator)

$$\hat{\beta}_{gmm} = \arg\min_{\beta} J(\beta) = \arg\min_{\beta} \; n\,\bar{g}_n(\beta)' W\,\bar{g}_n(\beta)$$

Theorem (13.1)

For the overidentified linear IV model, the GMM estimator is:

$$\hat{\beta}_{gmm} = (X'ZWZ'X)^{-1}(X'ZWZ'Y)$$

- When $\ell = k$ (just identified): $\hat{\beta}_{\text{gmm}} = (Z'X)^{-1}(Z'Y) = \hat{\beta}_{\text{iv}}$, regardless of W
- When $\ell > k$: the estimator depends on W

2SLS as GMM

Theorem (13.2)

If $W = (Z'Z)^{-1}$*, then* $\hat{\beta}_{gmm} = \hat{\beta}_{2sls}$*.*
Furthermore, if $k = \ell$ *then* $\hat{\beta}_{gmm} = \hat{\beta}_{iv}$*.*

Proof sketch: Substitute $W = (Z'Z)^{-1}$ into Theorem 13.1:

$$\hat{\beta} = (X'Z(Z'Z)^{-1}Z'X)^{-1}(X'Z(Z'Z)^{-1}Z'Y) = (X'P_ZX)^{-1}(X'P_ZY)$$

where $P_Z = Z(Z'Z)^{-1}Z'$ is the projection matrix — exactly the 2SLS formula.

Takeaway: 2SLS is a *one-step* GMM estimator. But $(Z'Z)^{-1}$ is not always the best weight matrix.

Asymptotic Distribution of GMM

Let $Q = \mathbb{E}[ZX']$ and $\Omega = \mathbb{E}[ZZ'e^2]$.

Theorem (13.3: Asymptotic Distribution of GMM)

As $n \to \infty$,

$$\sqrt{n}(\hat{\beta}_{gmm} - \beta) \xrightarrow{d} N(0, V_\beta)$$

where

$$\boxed{V_\beta = (Q'WQ)^{-1}(Q'W\Omega WQ)(Q'WQ)^{-1}}$$

- This is a **sandwich form**: "bread" $(Q'WQ)^{-1}$ wraps "meat" $Q'W\Omega WQ$
- Applies for *any* positive definite weight matrix W
- Different $W \Rightarrow$ different $V_\beta \Rightarrow$ different efficiency

What Is the Best Weighting?

Question: Which W minimizes V_β?

Intuition: We want to weight moments that are

- *more informative* (high signal) $\Rightarrow$ weight up
- *less noisy* (low variance) $\Rightarrow$ weight up

The variance of the sample moments is $\Omega = \mathbb{E}[g_i g_i']$. So weight inversely to their variance: $W = \Omega^{-1}$.

Analogy: Just like GLS weights observations by the inverse of $\text{Var}(e_i)$, efficient GMM weights *moment conditions* by the inverse of their covariance.

Efficient GMM

Theorem (13.4: Efficient GMM)

Setting $W = \Omega^{-1}$, *as* $n \to \infty$: $\sqrt{n}(\hat{\beta}_{gmm} - \beta) \xrightarrow{d} N(0, V_\beta)$ *where* $V_\beta = (Q'\Omega^{-1}Q)^{-1}$.

The sandwich collapses: $(Q'\Omega^{-1}Q)^{-1}(\underbrace{Q'\Omega^{-1}\Omega\Omega^{-1}Q}_{Q'\Omega^{-1}Q})(Q'\Omega^{-1}Q)^{-1}$

Theorem (13.5: Efficiency)

For any $W > 0$: $(Q'WQ)^{-1}(Q'W\Omega WQ)(Q'WQ)^{-1} - (Q'\Omega^{-1}Q)^{-1} \geq 0$.

No GMM estimator with these moment conditions can have a smaller asymptotic variance. Chamberlain (1987): this is the **semiparametric efficiency bound**.

When Is 2SLS Efficient?

Recall 2SLS uses $W = (Z'Z)^{-1}$, which converges to $(\mathbb{E}[ZZ'])^{-1}$.

The efficient weight is $\Omega^{-1} = (\mathbb{E}[ZZ'e^2])^{-1}$.

These are equal when $\mathbb{E}[e^2|Z] = \sigma^2$ (conditional homoskedasticity):

$$\mathbb{E}[ZZ'e^2] = \sigma^2\mathbb{E}[ZZ']$$

Theorem (13.6)

Under conditional homoskedasticity $\mathbb{E}[e^2|Z] = \sigma^2$, the 2SLS estimator $\hat{\beta}_{2sls}$ is efficient GMM.

Implications:

- Homoskedastic errors $\Rightarrow$ 2SLS is fine, no need for GMM
- Heteroskedastic errors $\Rightarrow$ 2SLS is *inefficient*; use efficient GMM

Two-Step GMM

Problem: $\Omega = \mathbb{E}[ZZ'e^2]$ is unknown. We need to estimate it.

Two-Step GMM:

1. **Step 1:** Estimate β by 2SLS (using $W = (Z'Z)^{-1}$). Compute residuals $\tilde{e}_i = Y_i - X_i'\hat{\beta}_{2sls}$.
2. **Step 2:** Estimate $\hat{\Omega} = \frac{1}{n}\sum_{i=1}^{n} Z_i Z_i' \tilde{e}_i^2$, set $\hat{W} = \hat{\Omega}^{-1}$, and re-estimate β.

Theorem (13.7)

The two-step GMM estimator is asymptotically efficient:

$$\sqrt{n}(\hat{\beta}_{gmm} - \beta) \xrightarrow{d} N(0, V_\beta) \quad \textit{where} \quad V_\beta = (Q'\Omega^{-1}Q)^{-1}$$

Key point: The initial estimator does not affect the asymptotic distribution.

Iterated GMM

The two-step estimator's *finite-sample* performance can depend on the initial estimator. To remove this dependence:

Iterated GMM:

1. Start with $\hat{\beta}^{(0)}$ (e.g., 2SLS)
2. Compute $\hat{\Omega}^{(s)} = \frac{1}{n}\sum Z_i Z_i'(\hat{e}_i^{(s)})^2$
3. Re-estimate: $\hat{\beta}^{(s+1)} = (X'Z\hat{\Omega}^{(s)-1}Z'X)^{-1}(X'Z\hat{\Omega}^{(s)-1}Z'Y)$
4. Repeat until convergence

In R's `gmm` package: `type = "twoStep"` (two-step) or `type = "iterative"` (iterated).

Note: Hansen and Lee (2021) show the iterated GMM estimator is unaffected by whether the weight matrix is computed with or without centering.

Covariance Matrix Estimation

One-step or two-step GMM (general W):

$$\hat{V}_\beta = (\hat{Q}'\hat{W}\hat{Q})^{-1}(\hat{Q}'\hat{W}\hat{\Omega}\hat{W}\hat{Q})(\hat{Q}'\hat{W}\hat{Q})^{-1}$$

where $\hat{Q} = \frac{1}{n}\sum Z_i X_i'$ and $\hat{\Omega}$ uses residuals from the final step.

Efficient iterated GMM ($\hat{W} = \hat{\Omega}^{-1}$):

$$\hat{V}_\beta = \left(\hat{Q}'\hat{\Omega}^{-1}\hat{Q}\right)^{-1}$$

The sandwich simplifies — just like GLS vs. OLS variance formulas.

Practical advice: Always report robust (sandwich) standard errors for one-step GMM. For efficient GMM, the simplified formula is already robust.

Application: Missing Data

Missing data on regressors is pervasive. Abrevaya & Donald (2017) surveyed top journals and found ∼50% of empirical papers have missing data; ∼70% simply drop incomplete observations.

Standard approaches:

1. **Complete case:** Drop missing obs.—consistent, but loses efficiency
2. **Dummy variable:** Fill in 0, add indicator—can be inconsistent
3. **Linear imputation:** Predict missing values—requires extra assumptions

The GMM approach: write down moment conditions for *both* complete and incomplete observations. The complete cases contribute structural moments; the incomplete cases contribute reduced-form moments. GMM combines them optimally.

The Missing Data Model (Abrevaya & Donald, 2017)

Structural regression:

$$y_i = \alpha_0 x_i + z_i'\beta_0 + \varepsilon_i \qquad \text{where } E(x_i\varepsilon_i) = 0,\ E(z_i\varepsilon_i) = 0$$

x_i is a (possibly missing) scalar regressor; z_i is always observed.

Linear projection of x_i on z_i: $x_i = z_i'\gamma_0 + \xi_i$ where $E(z_i\xi_i) = 0$. **Missingness indicator:** $m_i = 1$ if x_i missing, $m_i = 0$ if observed.

Key substitution: For observations with $m_i = 1$:

$$y_i = z_i'(\gamma_0\alpha_0 + \beta_0) + (\varepsilon_i + \xi_i\alpha_0) \equiv z_i'\delta_0 + \eta_i$$

Assumption 1: (a) $E(m_i z_i \varepsilon_i) = 0$; (b) $E(m_i z_i \xi_i) = 0$; (c) $E(m_i x_i \varepsilon_i) = 0$.
$\Rightarrow$ Missingness may depend on z_i but not on ε_i or ξ_i.

Three Blocks of Moment Conditions

The parameters are $\theta = (\alpha_0, \beta_0, \gamma_0)'$ with $k = 2K + 1$ unknowns.

Block 1 (observed cases, structural equation): $(1 - m_i) \cdot w_i(y_i - \alpha_0 x_i - z_i'\beta_0) = 0$ [$K + 1$ conditions]
Block 2 (observed cases, projection equation): $(1 - m_i) \cdot z_i(x_i - z_i'\gamma_0) = 0$ [K conditions]
Block 3 (missing cases, substituted equation): $m_i \cdot z_i(y_i - z_i'(\gamma_0\alpha_0 + \beta_0)) = 0$ [K conditions]

Total: $\ell = 3K + 1$ moments for $k = 2K + 1$ parameters $\Rightarrow$ K overidentifying restrictions (testable via J-test!)

With $K = 2$ (intercept + IQ): 7 moments, 5 parameters, **2 overidentifying restrictions**

Why GMM Beats the Alternatives

Method	**Consistent?**	**Efficient?**	**Testable?**
Complete case	Yes (drops $m_i = 1$)	No (fewer obs)	No
Dummy variable	Sometimes no	No	No
Linear imputation	Yes (under Assn 1)	Somewhat	No
GMM	**Yes** (under Assn 1)	**Yes** (optimal W)	**Yes** (J-test)

Abrevaya & Donald Monte Carlo ($n = 400$, 1000 replications):

Method	**Bias** (α_0)	$n \times$**Var** (α_0)	**MSE** (α_0)
Complete case	0.011	13.93	0.035
Dummy variable	-0.194	13.94	0.073
FGLS	0.011	13.93	0.035
GMM	**0.006**	**12.53**	**0.031**

WLS Data: Education and BMI

```
library(gmm); library(haven); library(estimatr)

# Wisconsin Longitudinal Study data
data1 <- read_dta("wls-data.dta")
data_men <- subset(data1, male == 1)

# How much data is missing?
table(data_men$bmimissing)
#   0    1
# 4231 1095    (~21% missing)

# Complete-case regression
data_complete <- subset(data_men, bmimissing == 0)
cc_reg <- lm(educ ~ bmirating + iq, data = data_complete)
```

Education (y) regressed on BMI rating (x, sometimes missing) and IQ (z, always observed).

Defining GMM Moment Functions in R

```
g_men <- function(theta, x) {
  beta_0 <- theta[1]; alpha <- theta[2]; beta_iq <- theta[3]
  gamma_0 <- theta[4]; gamma_iq <- theta[5]
  educ <- x[,1]; bmirating <- x[,2]; iq <- x[,3]; bmimissing <- x[,4]

  # Block 1: structural eq (observed cases)
  r1 <- (1 - bmimissing) * (educ - beta_0 - alpha*bmirating - beta_iq*iq)
  # Block 2: projection eq (observed cases)
  r2 <- (1 - bmimissing) * (bmirating - gamma_0 - gamma_iq*iq)
  # Block 3: substituted eq (missing cases)
  r3 <- bmimissing * (educ - (gamma_0*alpha+beta_0) - (gamma_iq*alpha+beta_iq)*iq)

  cbind(r1, r1*bmirating, r1*iq,  # 3 moments
        r2, r2*iq,                # 2 moments
        r3, r3*iq)                # 2 moments = 7 total, 5 params
}
```

Running GMM in R

```
# Starting values from complete-case regressions
start_men <- c(beta_0 = coef(cc_reg)[1],
               alpha  = coef(cc_reg)[2],
               beta_iq = coef(cc_reg)[3],
               gamma_0 = coef(proj_reg)[1],
               gamma_iq = coef(proj_reg)[2])

# Two-step GMM with HAC variance
gmm_men <- gmm(g_men, x = x_men, t0 = start_men,
                type = "twoStep",
                wmatrix = "ident", vcov = "HAC")
summary(gmm_men)
```

`type = "twoStep"`: initial $W = I$, then update with $\hat{\Omega}^{-1}$. `vcov = "HAC"`: HAC variance estimator.

Comparing Methods: Results

	Complete Case	Dummy Variable	GMM
$\hat{\alpha}$ (BMI)	-0.0343	-0.0343	-0.0340
SE	(0.0077)	(0.0073)	(0.0072)
$\hat{\beta}_{IQ}$	0.0563	0.0567	0.0564
SE	(0.0017)	(0.0015)	(0.0015)
n used	4231	5326	5326

J-test for overidentifying restrictions:

```
specTest(gmm_men)  # J-test = ?, df = 2, p-value = ?
```

- Large p-value: fail to reject $\Rightarrow$ Assumption 1 is plausible
- Small p-value: evidence against the linear projection restriction

Nonlinear GMM

So far: $g_i(\beta)$ is linear in β. In general, $g_i(\beta)$ can be **nonlinear**.

The GMM estimator still minimizes $J(\beta) = n\,\bar{g}_n(\beta)'\hat{W}\,\bar{g}_n(\beta)$ but now requires numerical optimization (no closed form).

Proposition 13.1: Distribution of Nonlinear GMM

$\sqrt{n}(\hat{\beta}_{\text{gmm}} - \beta) \xrightarrow{d} N(0, V_\beta)$ where $V_\beta = (Q'WQ)^{-1}(Q'W\Omega WQ)(Q'WQ)^{-1}$ with the **Jacobian** $Q = \mathbb{E}\left[\frac{\partial}{\partial \beta'} g_i(\beta)\right]$ replacing the linear $\mathbb{E}[ZX']$.

Example: The Abrevaya & Donald missing data model is nonlinear — Block 3 contains $\gamma_0\alpha_0$ (product of parameters).

The MTE Connection

Recall from Lecture 14: the **Marginal Treatment Effect** is

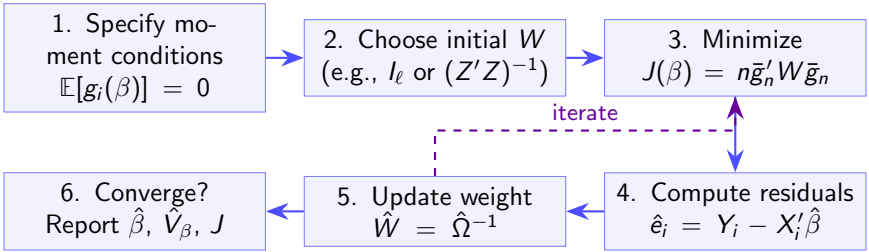
$$\Delta^{MTE}(x, u_D) = \mathbb{E}[Y_1 - Y_0 \mid X = x, U_D = u_D]$$

MTE estimation is nonlinear GMM:

- Moment conditions from the relationship between $E[Y \mid X, Z]$ and propensity score $P(Z)$
- MTE function modeled as a polynomial in u_D; parameters enter nonlinearly
- Overidentification arises from multiple instruments $\Rightarrow$ J-test applies

The `ivmte` package (Shea & Torgovitsky) implements this: specifying moment conditions, choosing weight matrices, and testing overidentifying restrictions.

Summary of GMM Estimation



- **One-step:** Stop after step 3 (e.g., 2SLS with $W = (Z'Z)^{-1}$)
- **Two-step:** One pass through steps 3–5
- **Iterated:** Repeat steps 3–5 until convergence

The OLS → GLS Parallel

The progression from 2SLS to efficient GMM mirrors Lectures 5–6:

	Linear model	**IV model**
Baseline	OLS	2SLS
Efficient under homoskedasticity?	Yes	Yes
Efficient under heteroskedasticity?	No ⇒ GLS	No ⇒ efficient GMM
Variance formula	$(X'\Omega^{-1}X)^{-1}$	$(Q'\Omega^{-1}Q)^{-1}$
Estimated how?	FGLS (two-step)	Two-step GMM

GMM is to IV what GLS is to OLS: use the covariance structure of the moments to improve efficiency.

Where We Stand

What GMM Buys You

1. **No distributional assumptions.** You specify moment conditions, not a likelihood—only orthogonality conditions.
2. **Optimal weighting.** $W = \Omega^{-1}$ gives the smallest asymptotic variance, paralleling GLS.
3. **Testable restrictions.** Extra moment conditions generate overidentifying restrictions you can test with the J-statistic (next lecture).

Every estimator in this course—OLS, GLS, IV, 2SLS—is a special case. GMM makes explicit what each one assumes and what it trades off.

Preview of Next Lecture

Lecture 16: GMM Inference and Going Beyond ATE

- **Testing:** The J-test for overidentification (Thm 13.14)
- **Hypothesis tests:** Wald test and Distance test in the GMM framework
- **Subset tests:** Testing specific instruments, endogeneity tests
- **Application: Propaganda and Foreign Media**
  - Kern & Hainmueller: West German TV in East Germany
  - IV gives LATE ≈ -0.12 (media reduces support for regime)
  - MTE reveals treatment effect heterogeneity: always-takers vs. never-takers have *opposite signs*
- **The course in one slide:** OLS $\rightarrow$ GLS $\rightarrow$ IV $\rightarrow$ 2SLS $\rightarrow$ GMM